7.6节
轮系的运动分析和设计

授课提纲

1. 知识回顾：齿轮传动类型
2. 问题导入：大传动比实现
   分路传送

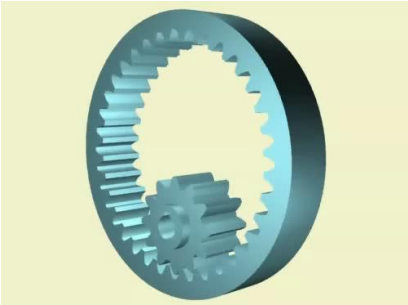
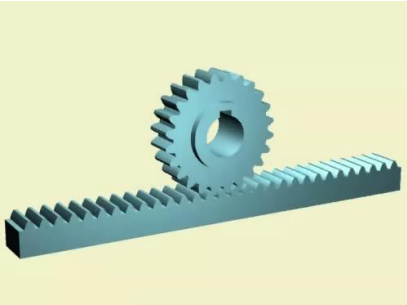
3. 设计鉴赏：指南车-历史机械智慧
   机械表-现代机械之美

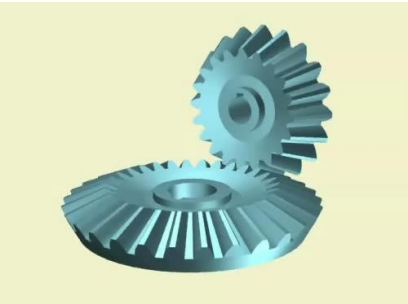
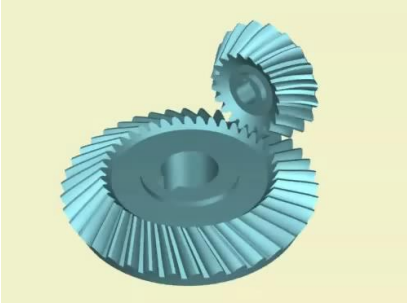
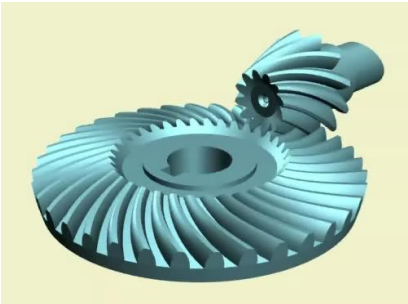
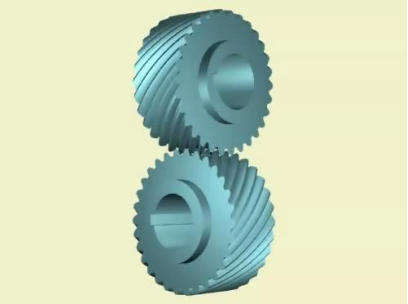
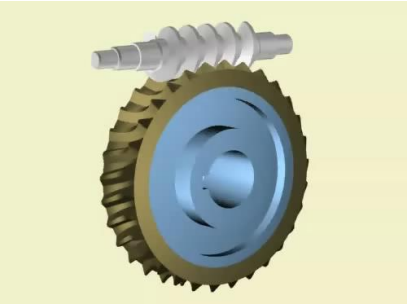
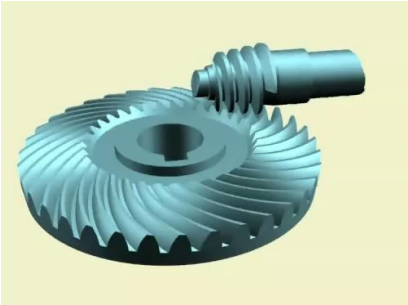
4. 轮系定义与分类

5. 轮系传动比计算

   定轴轮系传动比计算
   周转轮系传动比计算
   混合轮系传动比计算

1 知识回顾：齿轮传动类型

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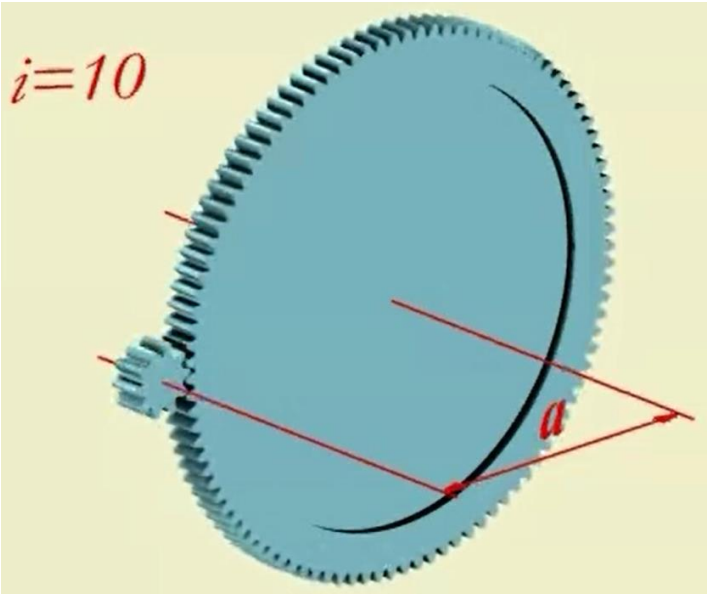
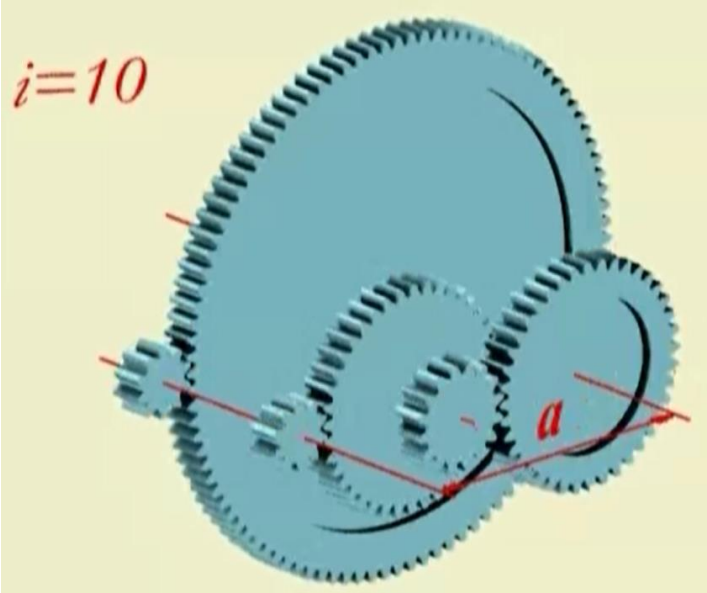
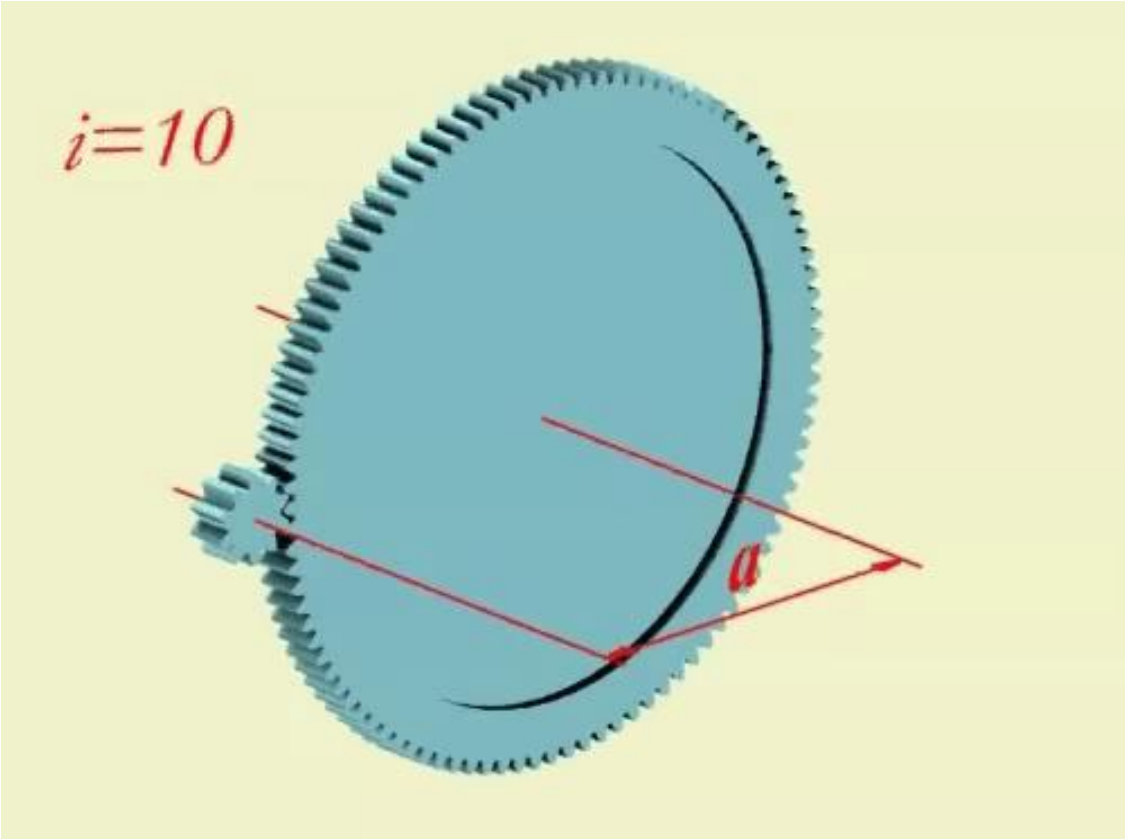
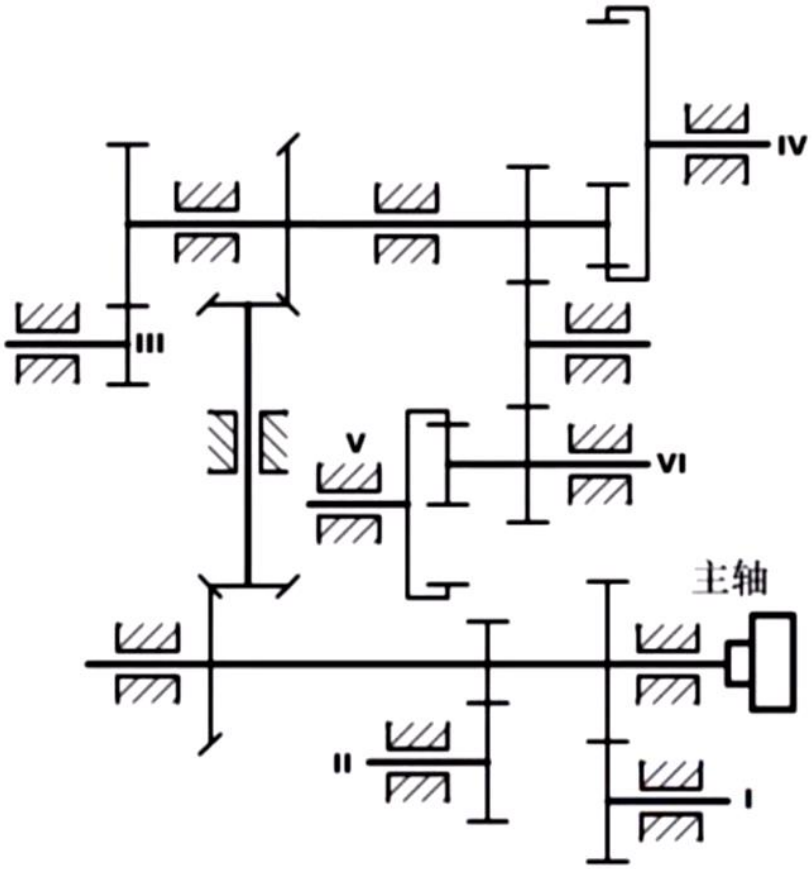
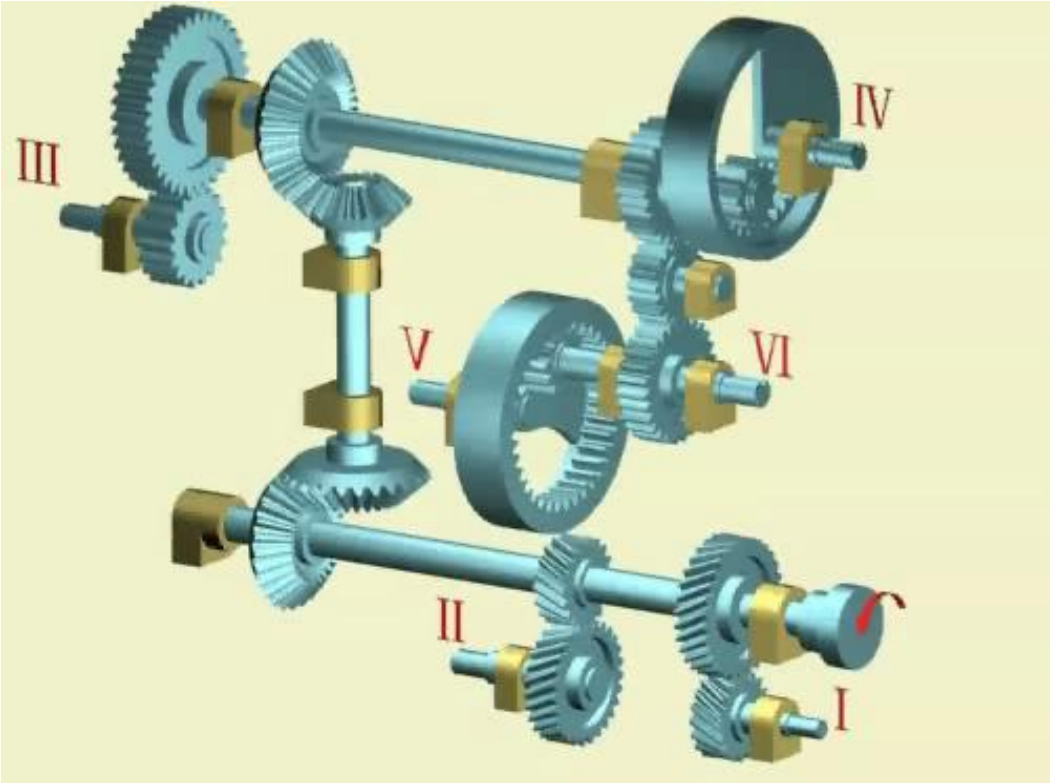




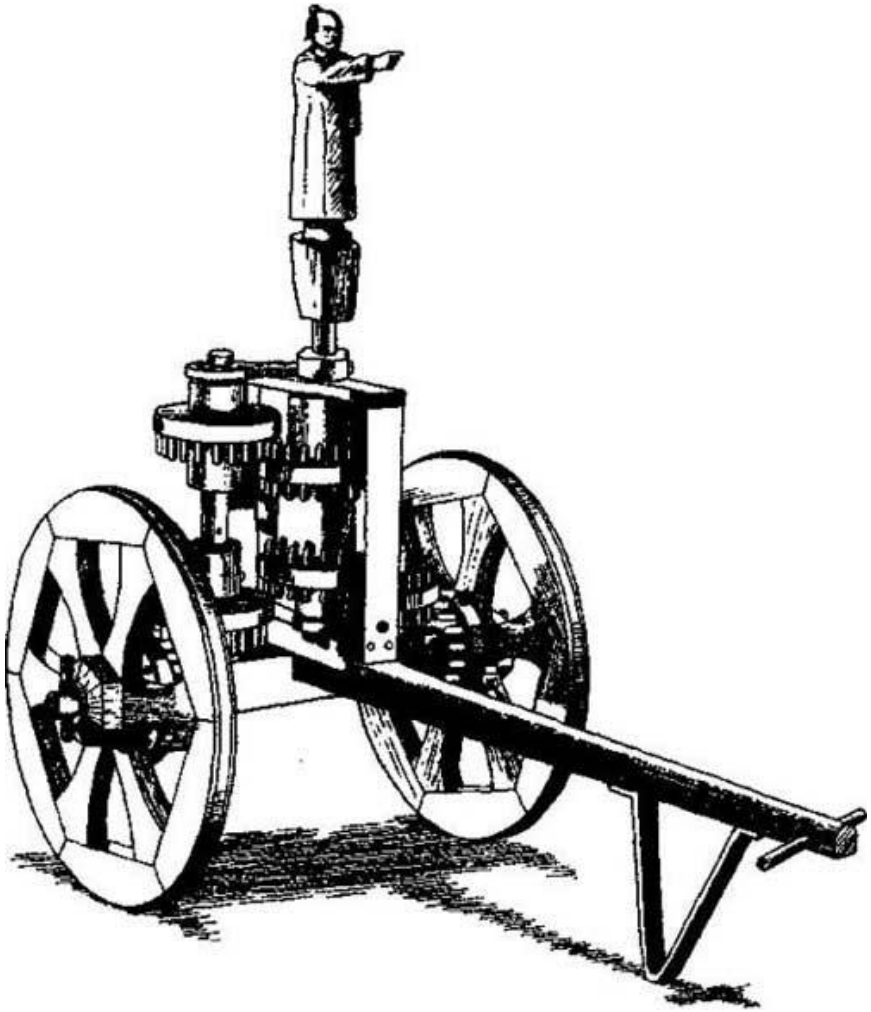


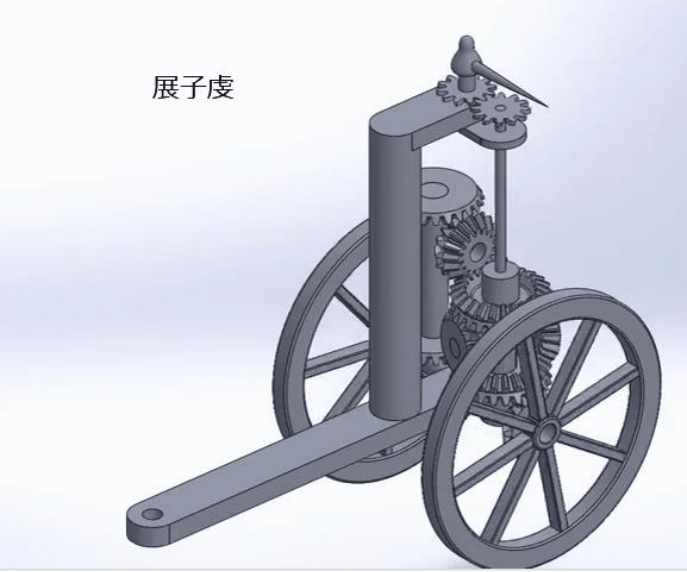
2 问题导入：分路传送

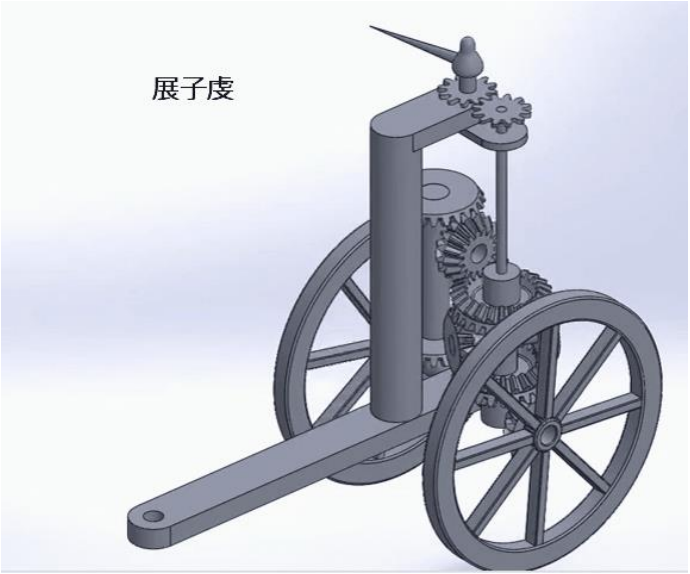




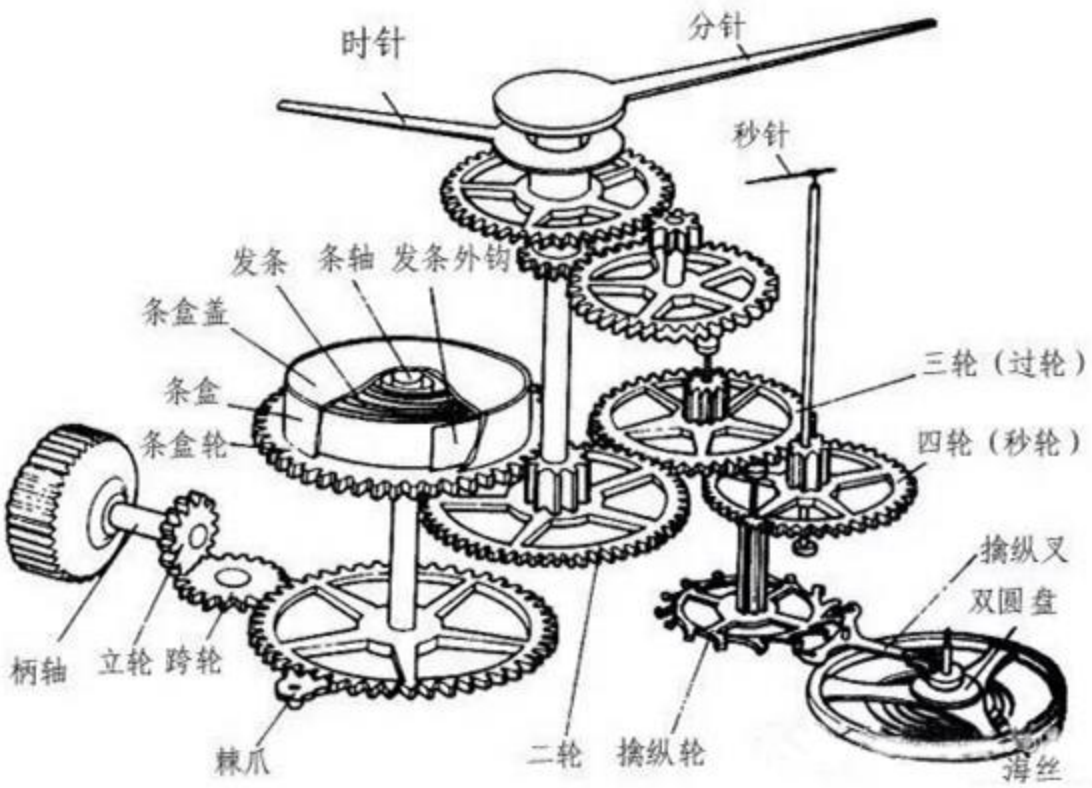
3 鉴赏1：指南车-古人机械智慧







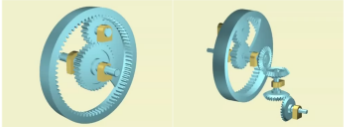
3 鉴赏2：机械表-现代机械之美



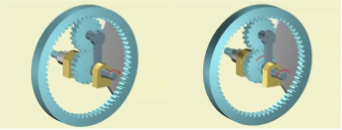
4 轮系定义与分类

轮系： 由一系列互相啮合的齿轮组成的传动机构，用于**原动机和执行机构之间的运动和动力传递.**

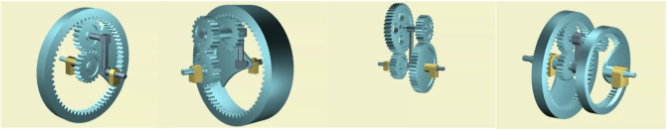
- 轮系分类
  - 定轴轮系
    - 平面定轴轮系
    - 空间定轴轮系



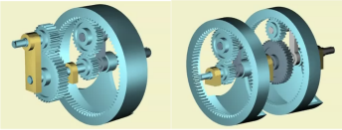
  - 周转轮系
    - 根据自由度数目
      - 行星轮系
      - 差动轮系



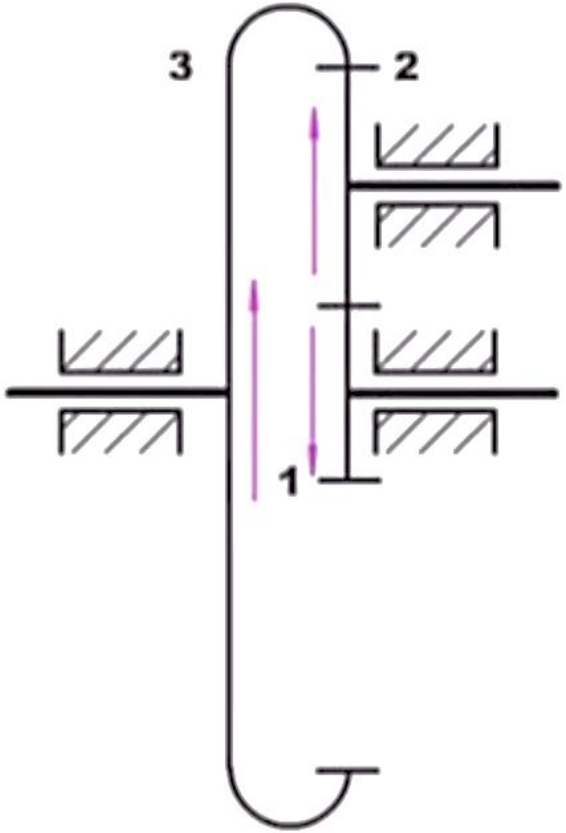
    - 根据基本构件特点
      - 2K-H型
      - 3K-H型

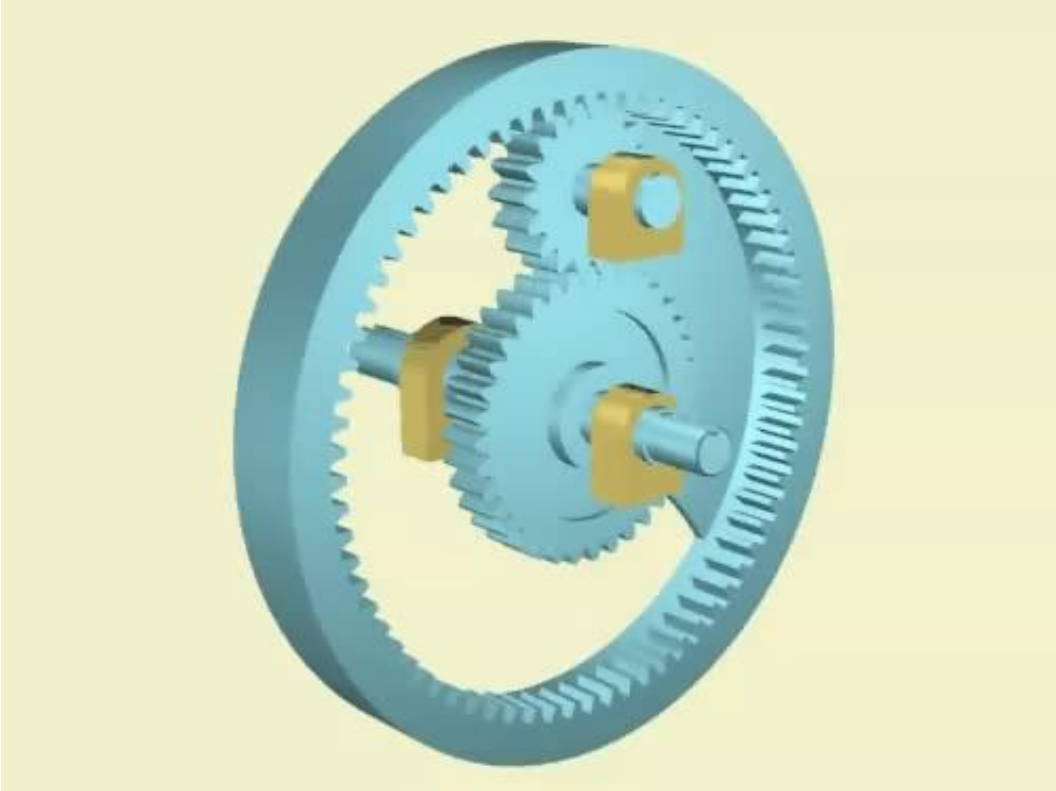


  - 混合轮系
    - 定轴轮系+周转轮系
    - 周转轮系+周转轮系



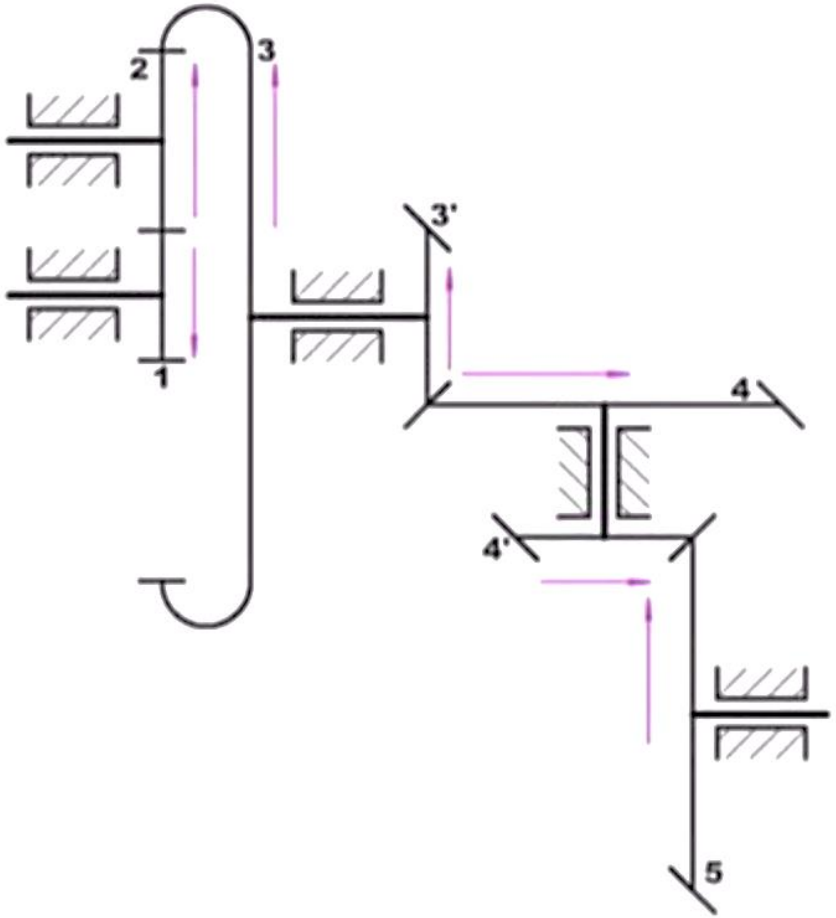
定轴轮系（平面定轴轮系）

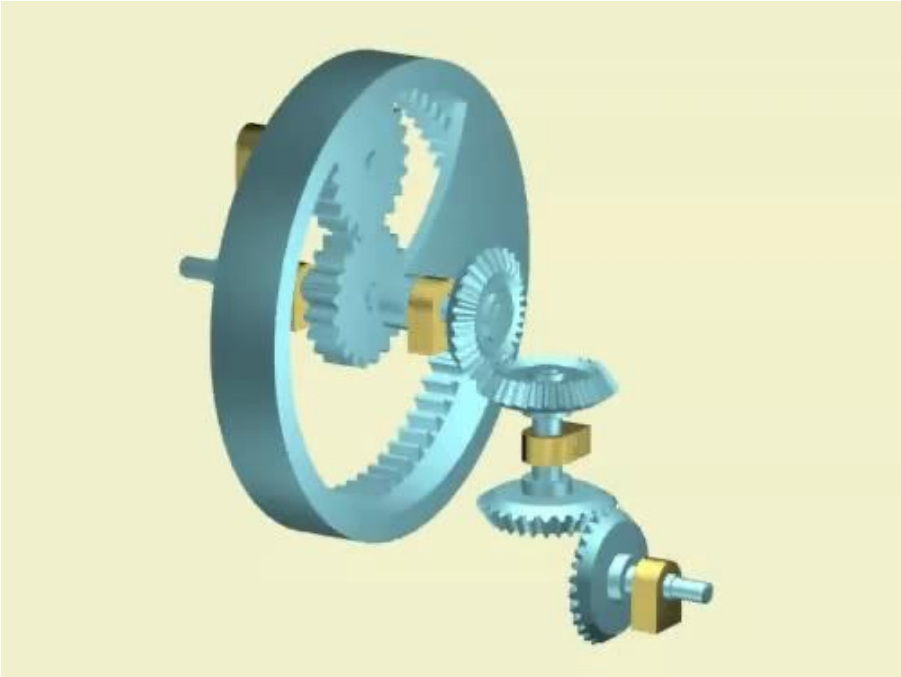




平面定轴轮系：各齿轮轴心线相互平行或重合

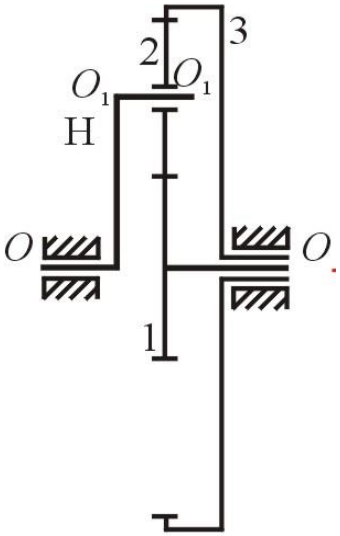
定轴轮系（空间定轴轮系）

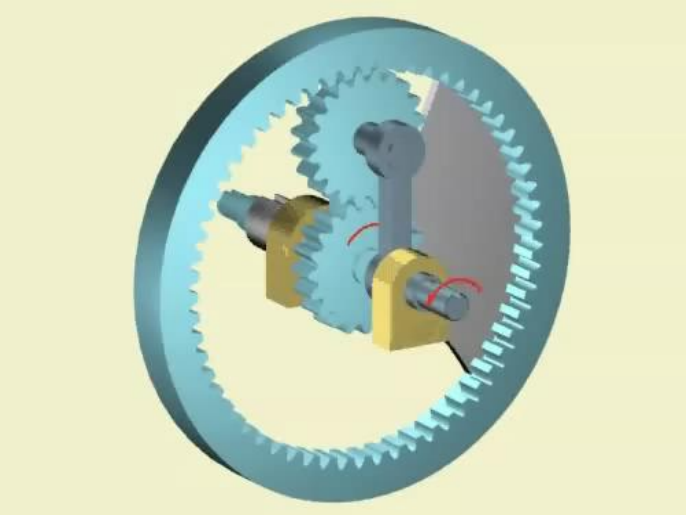




空间定轴轮系：某些齿轮轴心线相互不平行或不重合

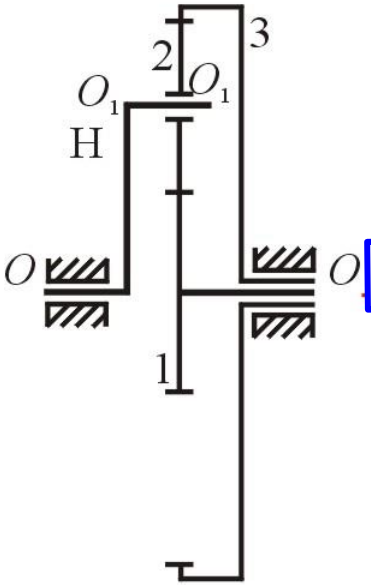
周转轮系（基本构件）





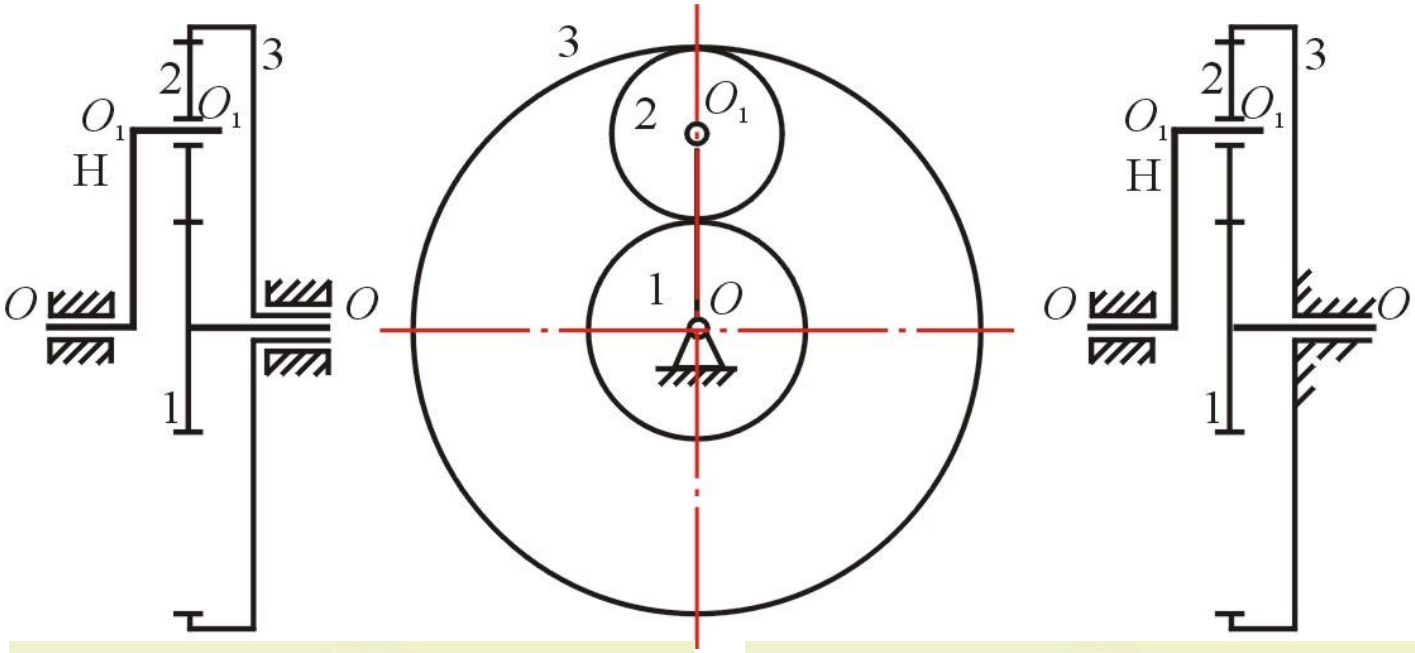
基本构件

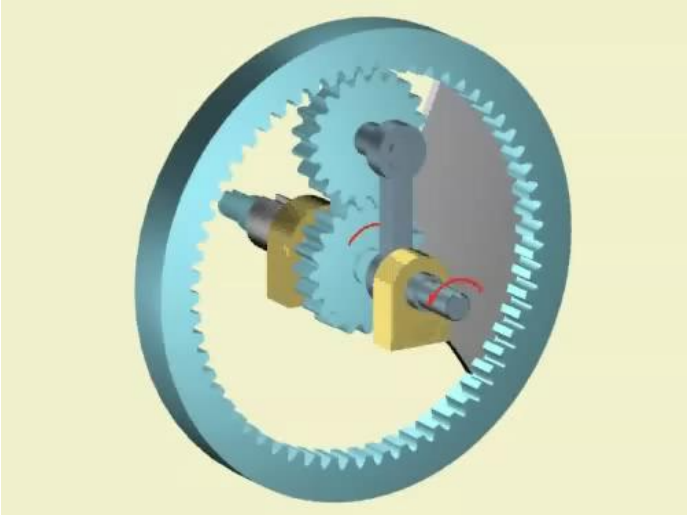
- 2-行星轮
- H-系杆
- 1-中心轮
- 3-中心轮



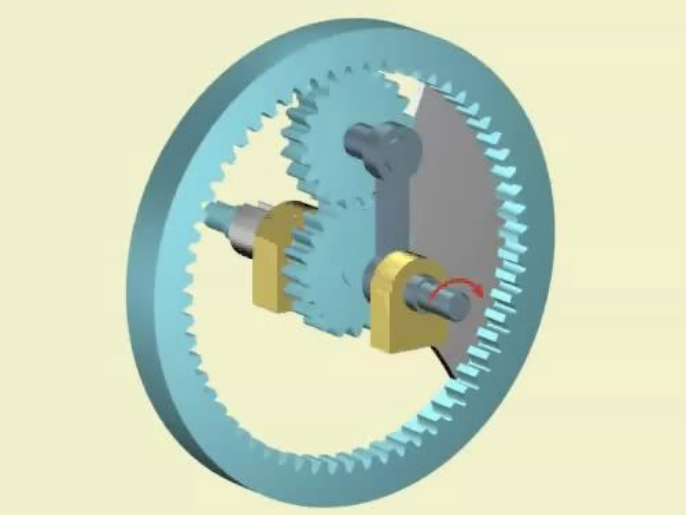
基本构件都是围绕着同一固定轴线回转的

周转轮系（按自由度数目分）



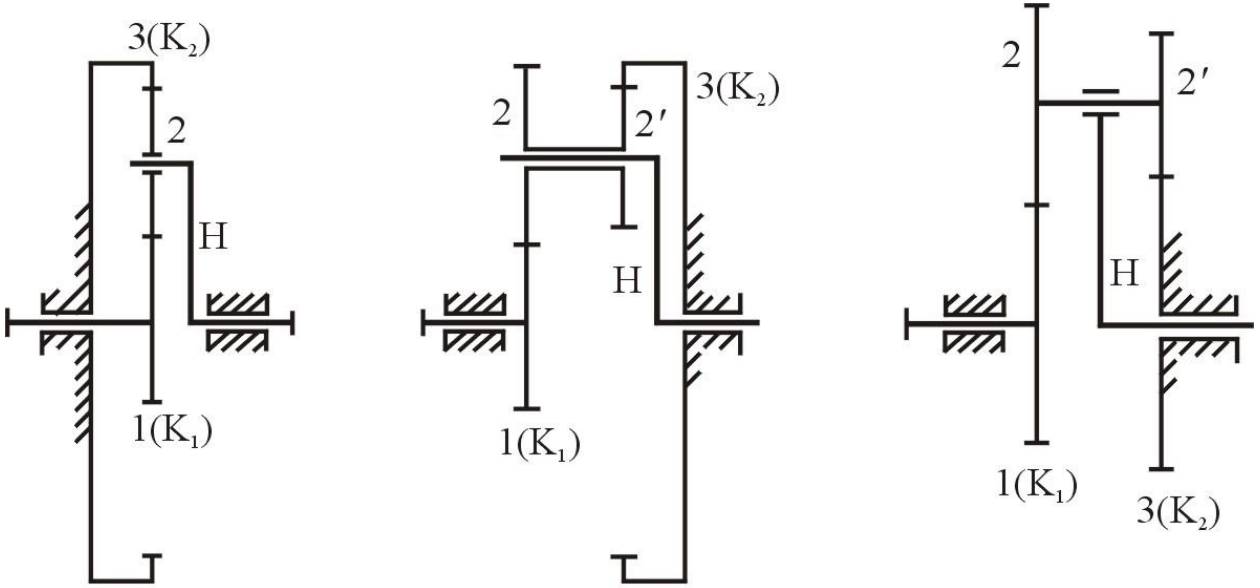


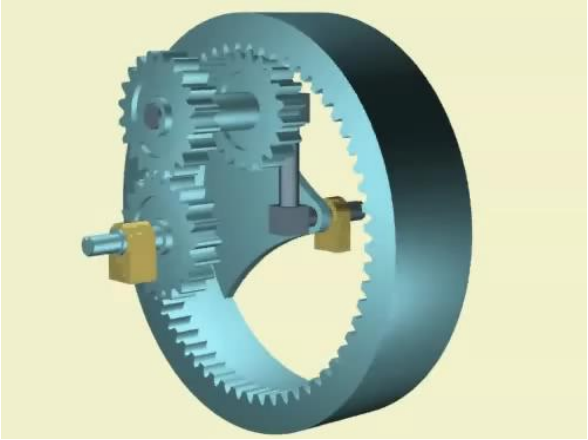
差动轮系(F=2)

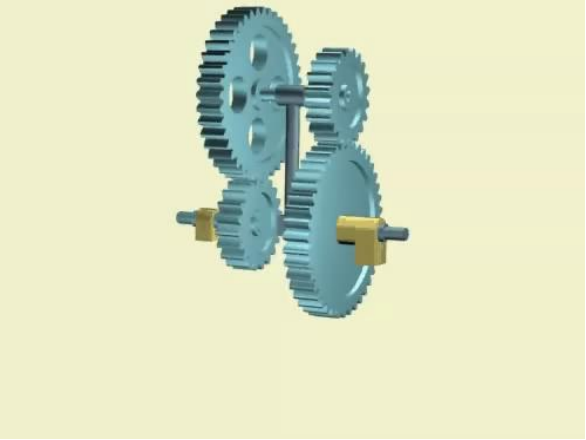


行星轮系(F=1)

周转轮系（按基本构件特点分）



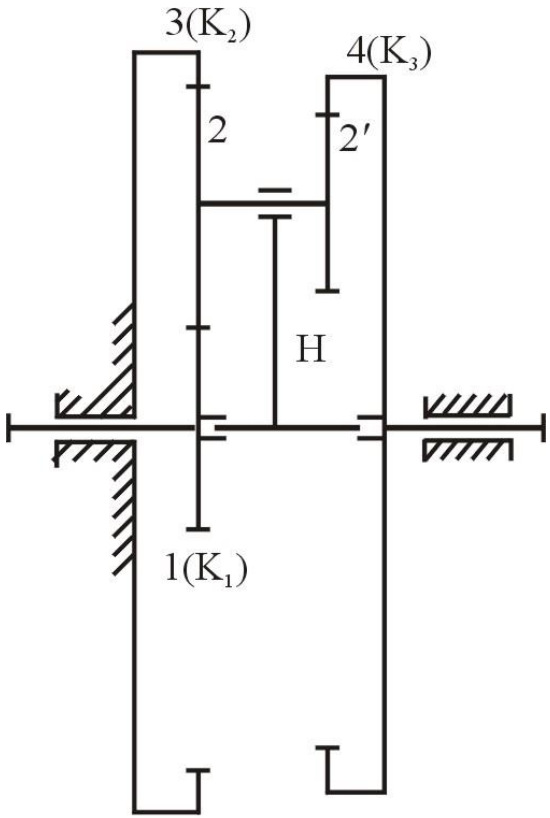


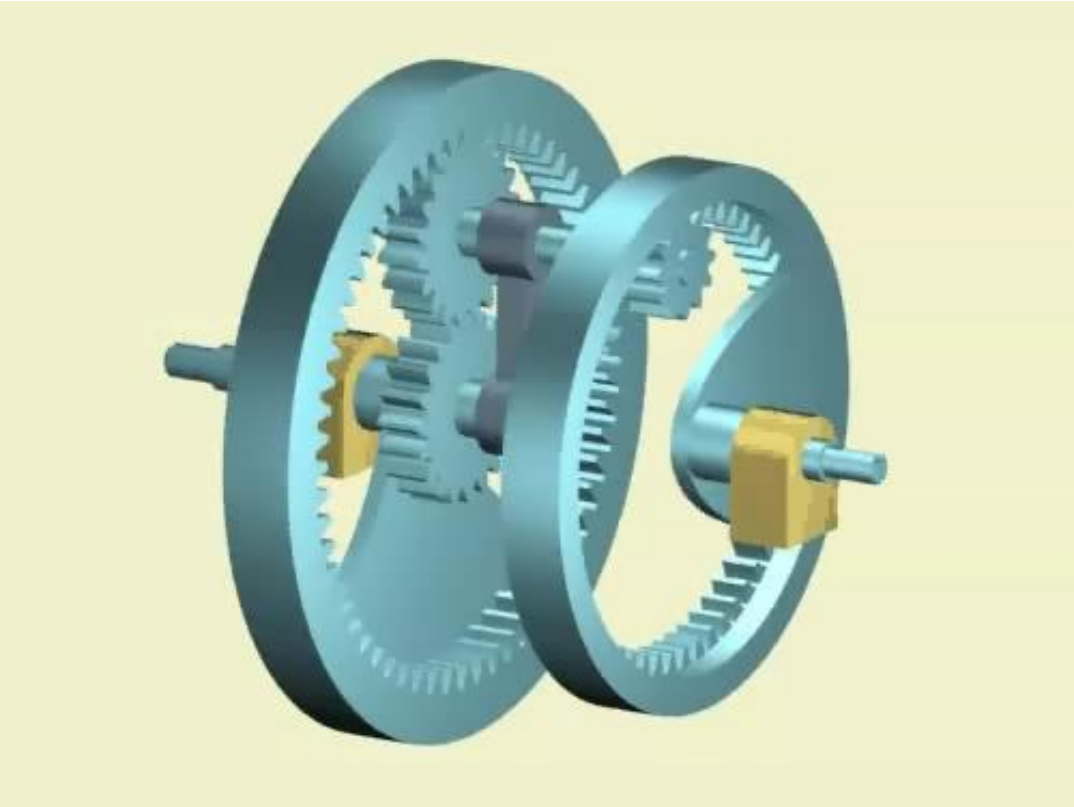


2K-H型

(周转轮系中包含2个太阳轮)

周转轮系（按基本构件特点分）

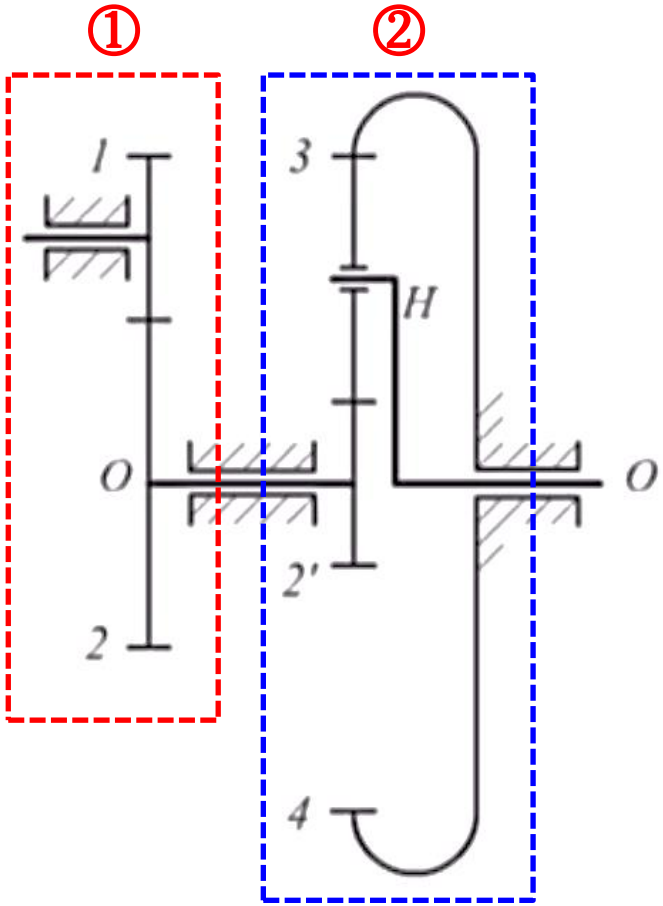


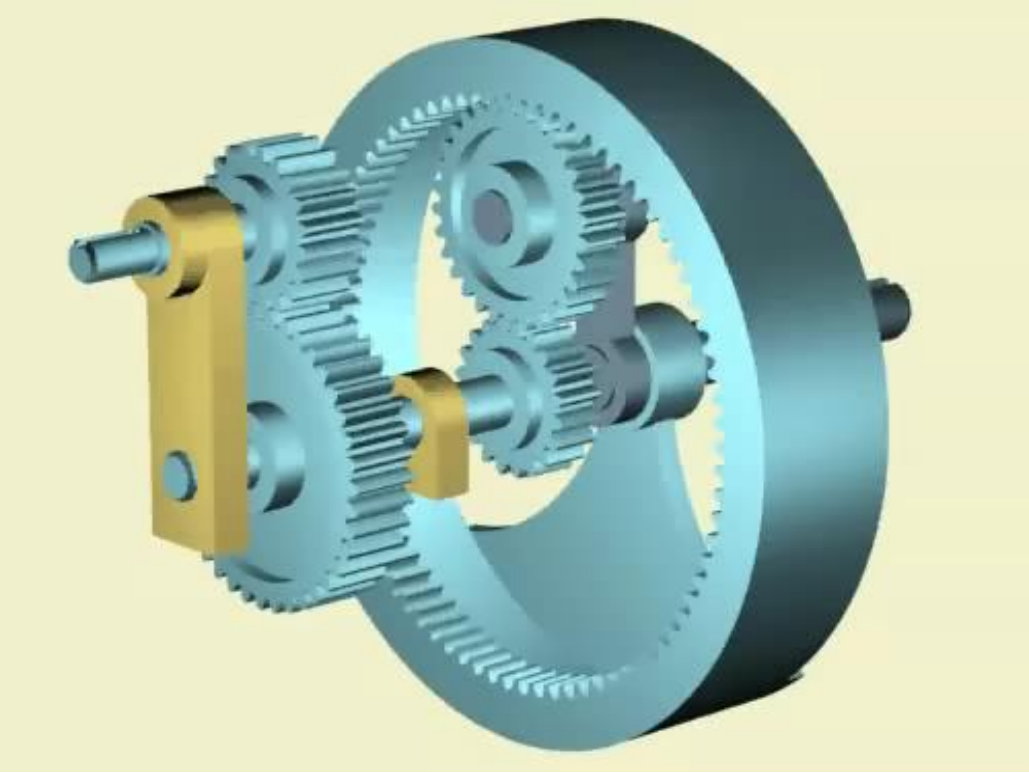


3K-H型

(周转轮系中包含3个太阳轮，系杆H只起到支撑行星轮的作用，不作为运动输入或输出构件)

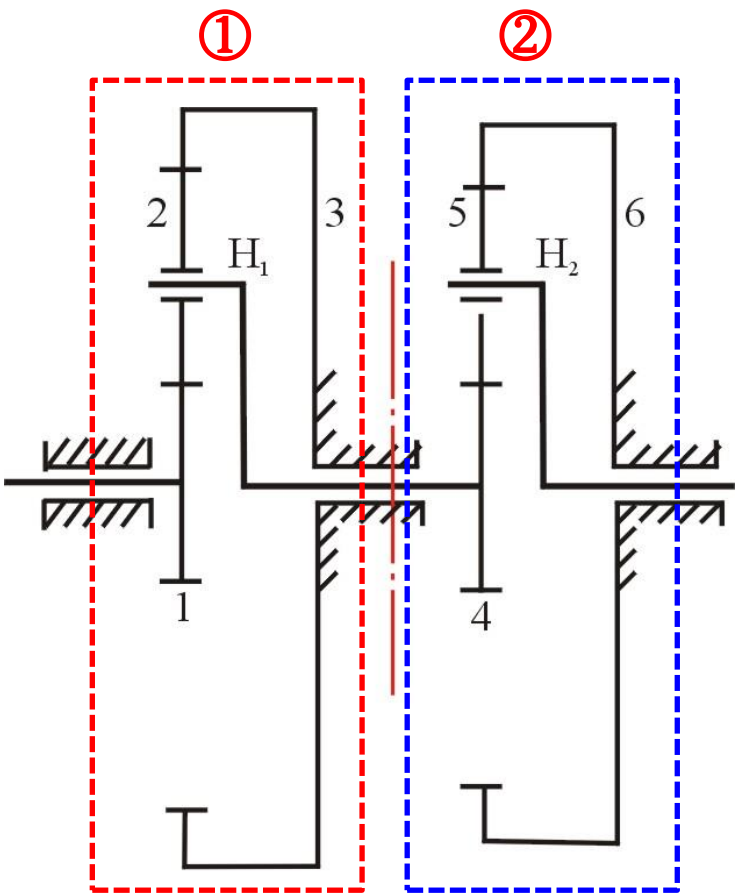
复合轮系（定轴+周转）





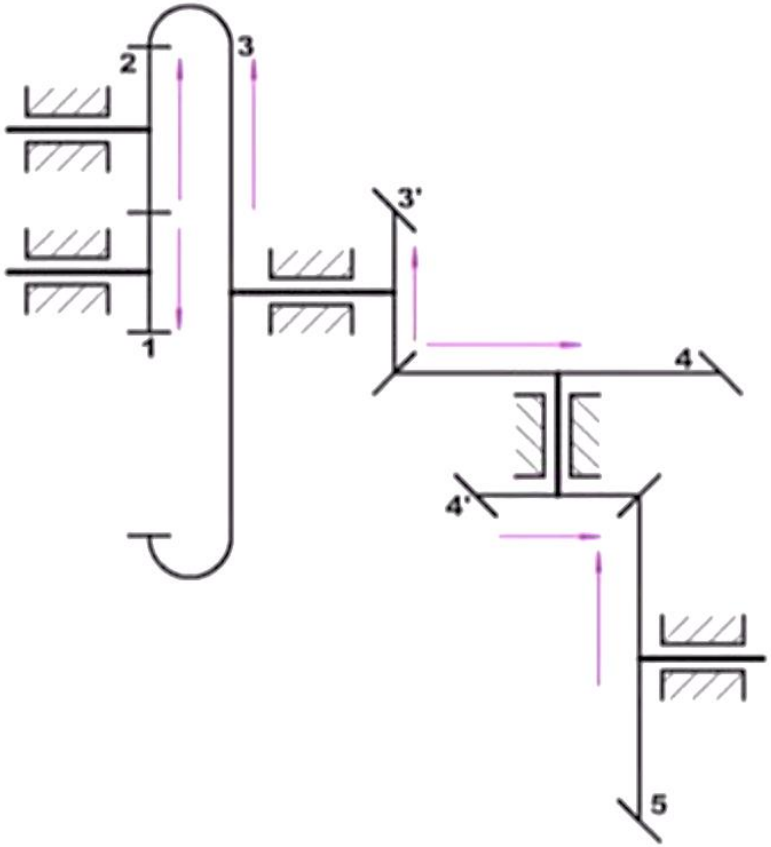
定轴轮系①+周转轮系②

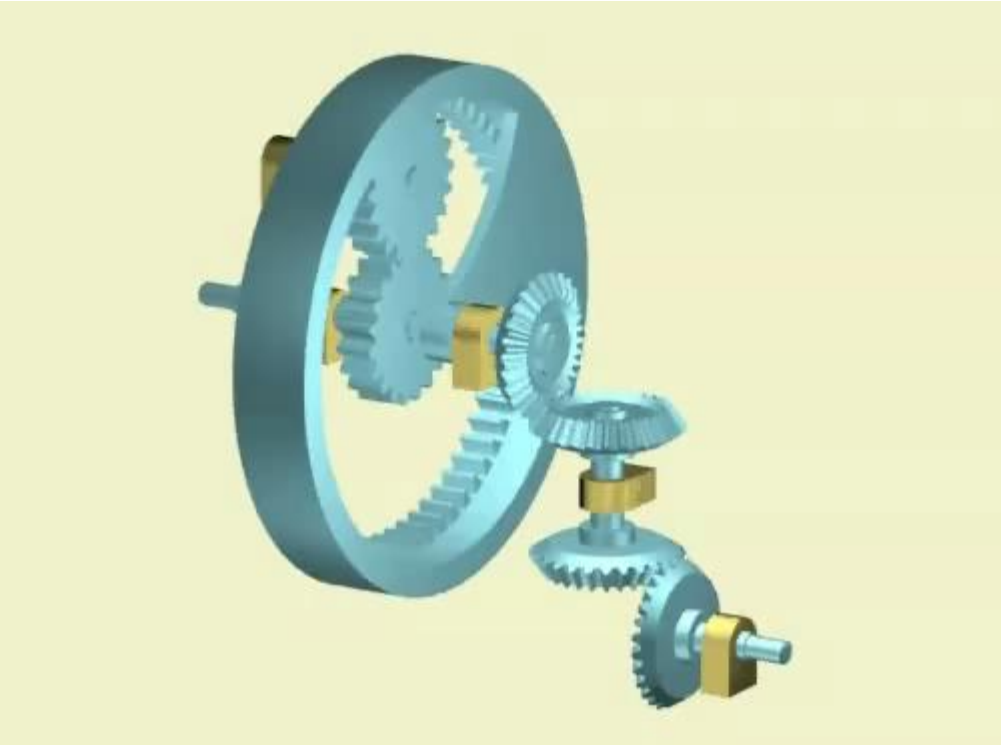
复合轮系（周转+周转）



周转轮系①+周转轮系②

5 定轴轮系传动比计算





$$i_{1k} = \frac{\omega_1}{\omega_k}$$

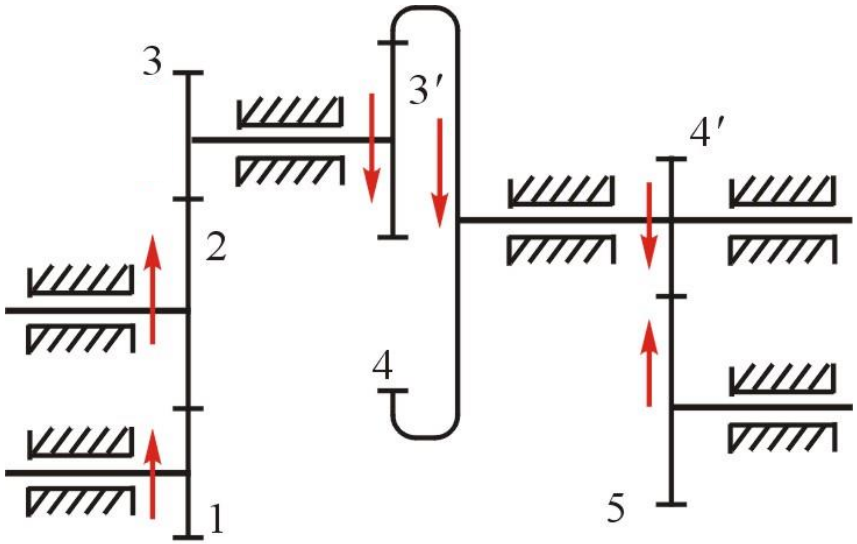
①传动比大小+②转向关系

定轴轮系传动比计算（传动比大小）

$$i_{15}=\frac{\omega_1}{\omega_5}=?$$

$$i_{12}=\frac{\omega_1}{\omega_2}=\frac{z_2}{z_1}\qquad i_{23}=\frac{\omega_2}{\omega_3}=\frac{z_3}{z_2}$$

$$i_{3'4}=\frac{\omega_3}{\omega_4}=\frac{z_4}{z_{3'}}\qquad i_{4'5}=\frac{\omega_4}{\omega_5}=\frac{z_5}{z_{4'}}$$



$$i_{15}=\frac{\omega_1}{\omega_5}=i_{12}\cdot i_{23}\cdot i_{3'4}\cdot i_{4'5}=\frac{\omega_1}{\omega_2}\cdot\frac{\omega_2}{\omega_3}\cdot\frac{\omega_3}{\omega_4}\cdot\frac{\omega_4}{\omega_5}$$

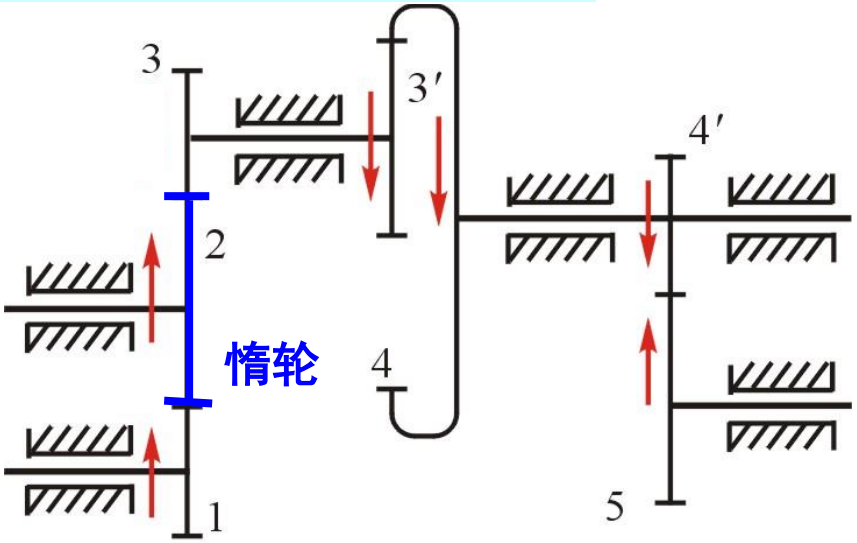
$$=\frac{z_2z_3z_4z_5}{z_1z_{2'}z_{3'}z_4}=\frac{\text{所有从动轮齿数的乘积}}{\text{所有主动轮齿数的乘积}}$$

定轴轮系传动比计算（转向关系）

A.平面定轴轮系（各齿轮轴线相互平行）

$$i_{15}=\frac{\omega_1}{\omega_5}=(-1)^3\frac{z_2z_3z_4z_5}{z_1z_{2'}z_{3'}z_4}$$

$$=-\frac{z_2z_3z_4z_5}{z_1z_{2'}z_{3'}z_4}$$

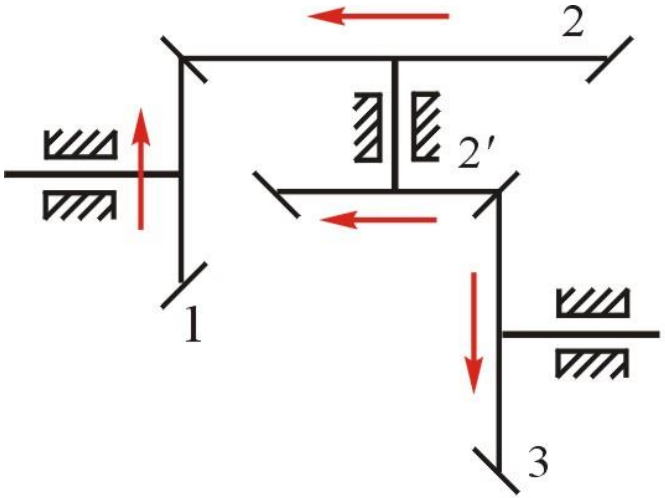


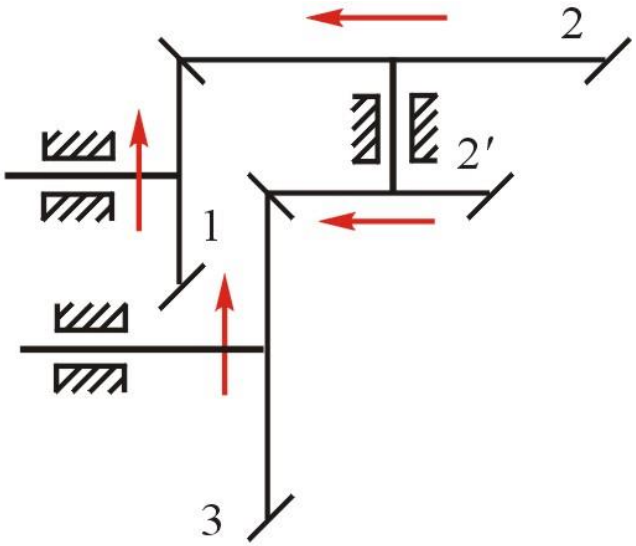
$$i_{1k}=\frac{\omega_1}{\omega_k}=\boxed{(-1)^m}\frac{\text{所有从动轮齿数的乘积}}{\text{所有主动轮齿数的乘积}}$$

惰轮:不改变传动比的大小，但改变轮系的转向

定轴轮系传动比计算（转向关系）

B. 定轴轮系中各轮几何轴线不都平行，但是输入、输出轮的轴线相互平行的情况





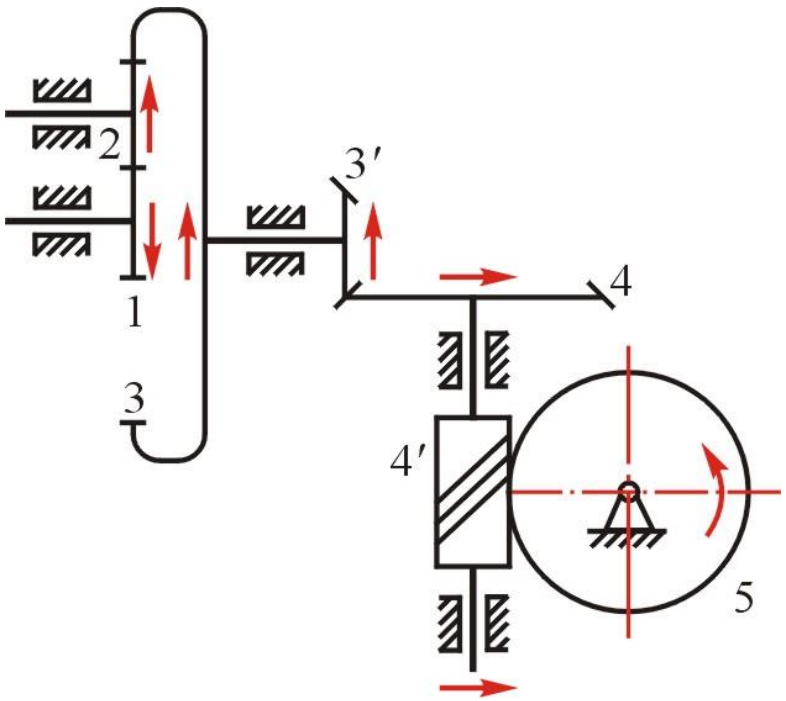
传动比方向判断 ⇨ 画箭头

传动比方向表示 ⇨ 在传动比的前面加正、负号

定轴轮系传动比计算（转向关系）

C. 输入、输出轮的轴线不平行的情况

齿轮1的轴为**输入轴**，蜗轮5的轴为**输出轴**，输出轴与输入轴的转向关系如图上箭头所示。

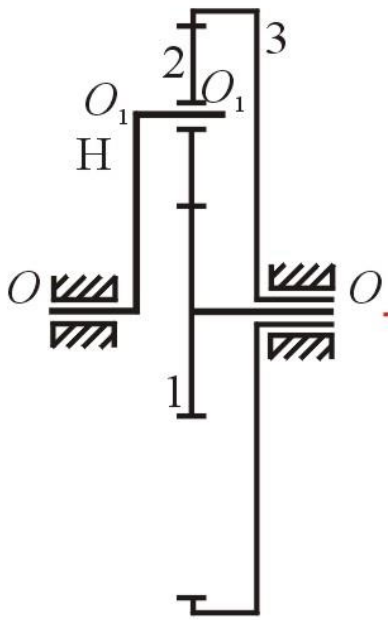


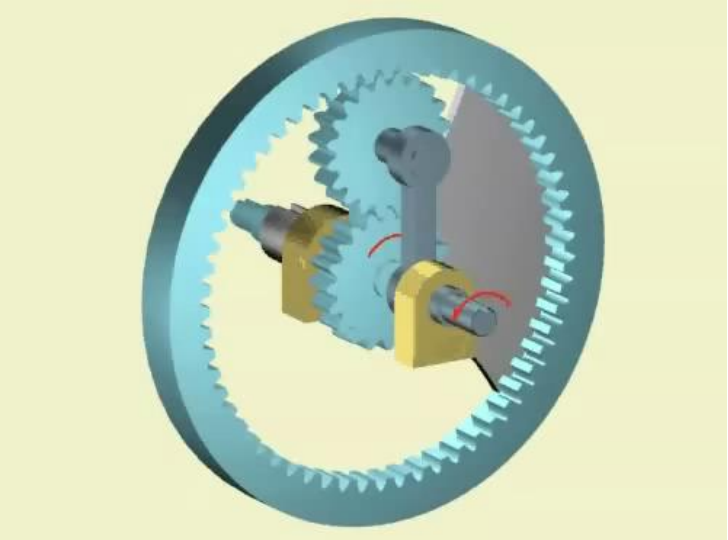
$$i_{15} = \frac{z_2 z_3 z_5}{z_1 z_{2'} z_{3'}}$$

传动比方向判断 ⇨ 画箭头

传动比方向表示

6 周转轮系传动比计算





定轴轮系传动比计算公式

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周转轮系传动比计算

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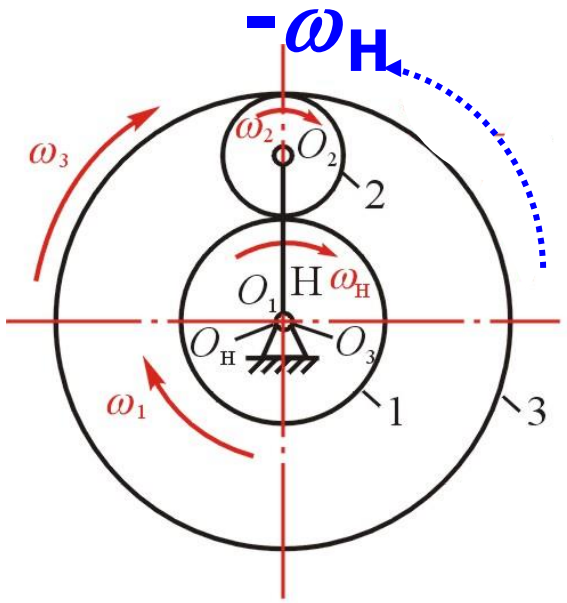
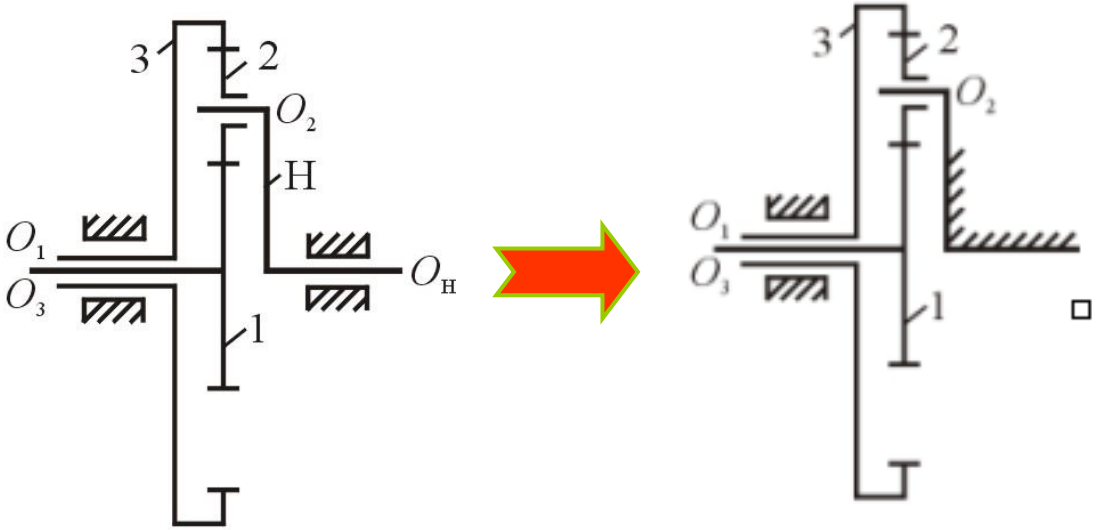
反转法原理

将周转轮系转化为定轴轮系

周转轮系传动比计算（计算思路）

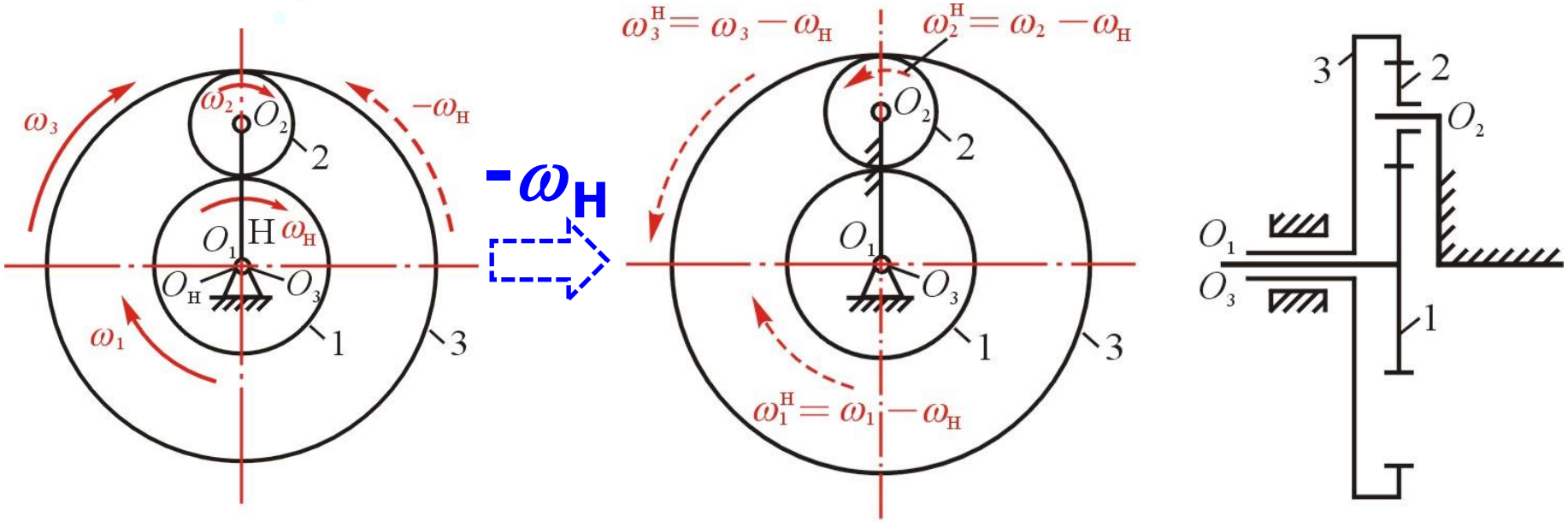




系杆➔机架

周转轮系➔定轴轮系

周转轮系传动比计算（计算方法）



$$i_{13}^{H} = \frac{\omega_1^H}{\omega_3^H} = \frac{\omega_1 - \omega_H}{\omega_3 - \omega_H} = (-1)^1 \cdot \frac{Z_3}{Z_1}$$

上式“－”说明在转化轮系中ω^H_1与ω^H_3方向相反。

周转轮系传动比计算（结果讨论）

一般周转轮系转化机构的传动比

$$i_{1K}^{H}=\frac{\omega_1^{H}}{\omega_K^{H}}=\frac{\omega_1-\omega_H}{\omega_K-\omega_H}=\pm\frac{z_2\cdots z_K}{z_1\cdots z_{K-1}}$$

A. 对于差动轮系: 给定w_1、w_k、w_H中的任意两个，可以计算出第三个，从而可以计算周转轮系的传动比。

周转轮系传动比计算（结果讨论）

B. 对于行星轮系: 两个中心轮中必有一个是固定的

若 $\omega_{\mathrm{K}} = 0$

$$i_{1k}^{\mathrm{H}} = \frac{\omega_1^{\mathrm{H}}}{\omega_{\mathrm{K}}^{\mathrm{H}}} = \frac{\omega_1 - \omega_{\mathrm{H}}}{0 - \omega_{\mathrm{H}}} = 1 - \frac{\omega_1}{\omega_{\mathrm{H}}} = 1 - i_{1\mathrm{H}} = \pm \frac{z_2 \cdots z_{\mathrm{K}}}{z_1 \cdots z_{\mathrm{K}-1}}$$

$$i_{1\mathrm{H}} = 1 - i_{1\mathrm{K}}^{\mathrm{H}}$$

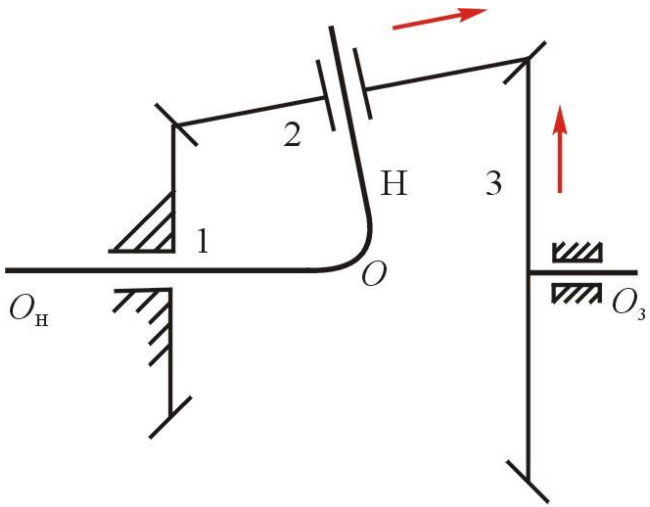
如果给定另外两个基本构件的角速度w_1、w_{H}中的任意一个，可以计算出另外一个，从而可以计算周转轮系的传动比。

周转轮系传动比计算（注意事项）

A. 转化轮系的1轮、k轮和系杆H的轴线需平行

$$i_{13}^{\mathrm{H}}=\frac{\omega_1-\omega_{\mathrm{H}}}{\omega_3-\omega_{\mathrm{H}}}=(-1)^2\frac{Z_2Z_3}{Z_1Z_2}$$

$$i_{12}^{\mathrm{H}}\neq\frac{\omega_1-\omega_{\mathrm{H}}}{\omega_2-\omega_{\mathrm{H}}}$$ （需作矢量作）



B. i_{1k}^{H} 是转化机构中1为主动轮、k为从动轮时的传动比，其大小和正、负完全按照根据定轴轮系来处理。周转轮系传动比正负是计算出来的，而不是判断出来的。

C. 表达式中 w_1、w_k、w_H的正负号问题。若基本构件的实际转速方向相反，则 w 的正负号应该不同。

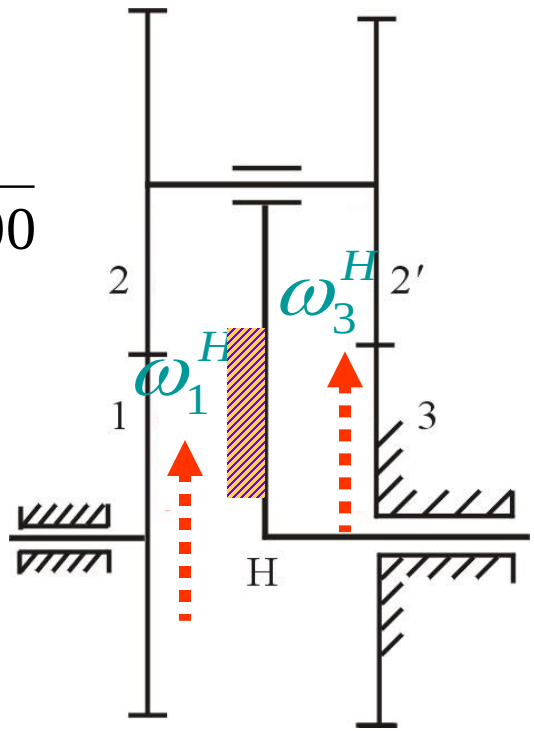
周转轮系传动比计算（计算举例）

例题1： 已知 $z_1=100$，$z_2=101$，$z_2'=100$，$z_3=99$，

试求传动比。

$$i_{1H}=1-i_{13}^{H}=1-\frac{z_2z_3}{z_1z_2'}=1-\frac{101\times 99}{100\times 100}=\frac{1}{10000}$$

$$i_{H1}=1/i_{1H}=10000$$



当系杆转10000转时，轮1才转1转，二者转向相同。此例说明周转轮系可获得很大的传动比。

周转轮系传动比计算（计算举例）

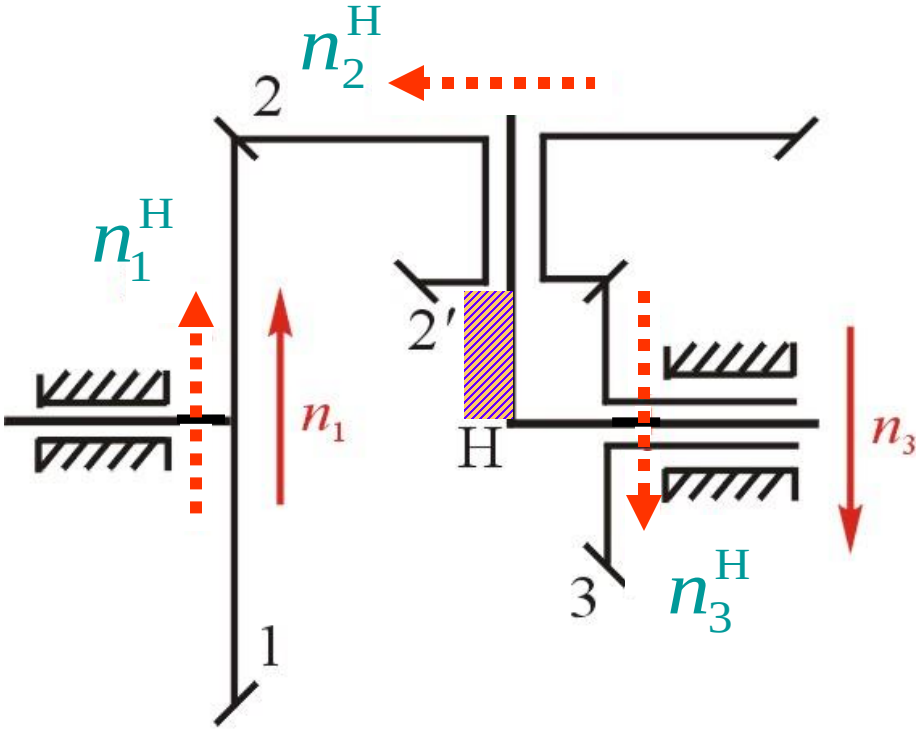
例题2： $z_1=z_2=48$，$z_{2'}=18$，$z_3=24$，$n_1=250$ r/min，$n_3=100$ r/min，方向如图所示。求：n_H 的大小和方向

$$i_{13}^{H}=\frac{n_1^H}{n_3^H}=\frac{n_1-n_H}{n_3-n_H}=-\frac{z_2z_3}{z_1z_{2'}}$$

$$=-\frac{48\times24}{48\times18}=-\frac{4}{3}$$

$$\frac{250-\omega_H}{-100-\omega_H}=-\frac{4}{3}$$

$$\omega_H=+50$$

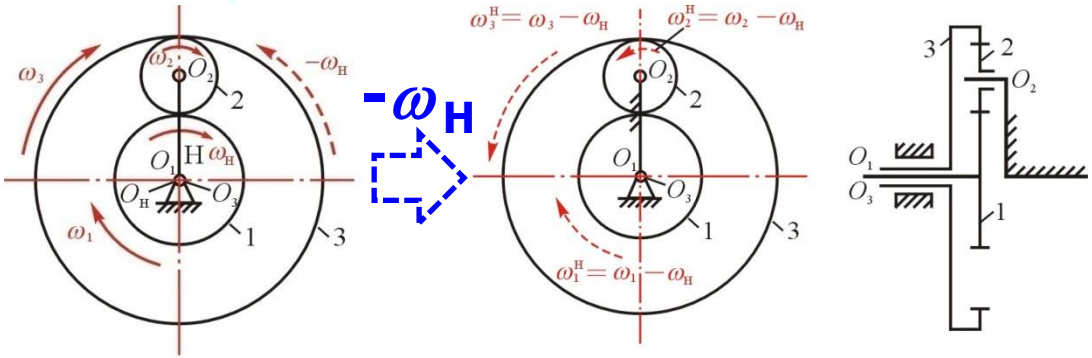


周转轮系传动比计算（小结）

周转轮系

$-\omega_H$

转化机构：假想的定轴轮系



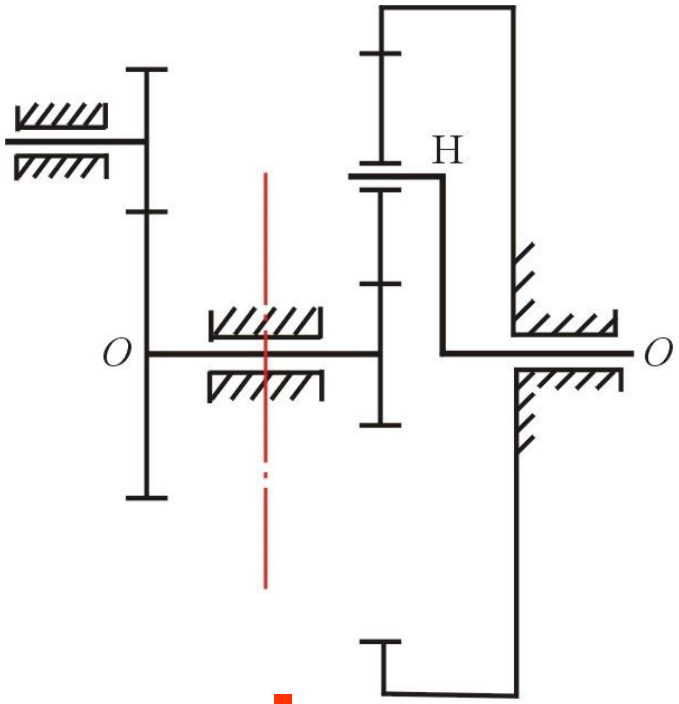
计算转化机构的传动比

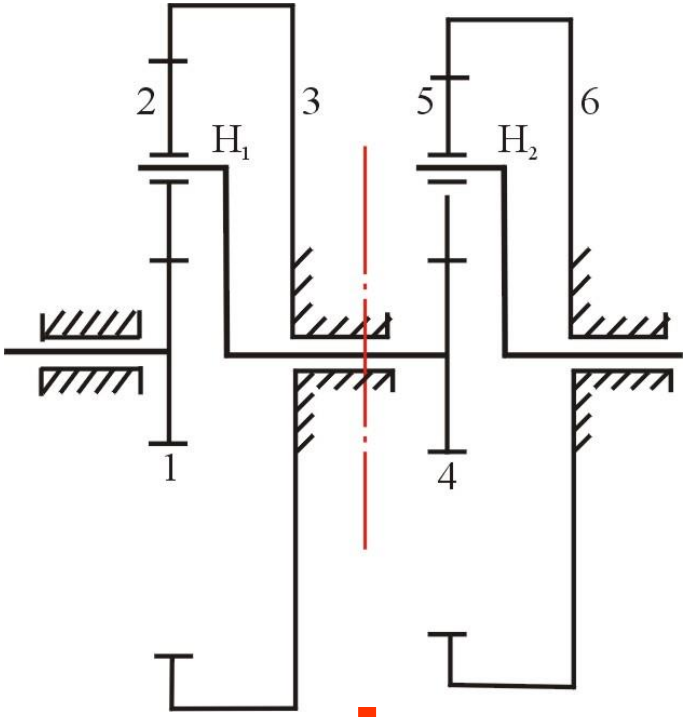
$$i_{1k}^{H} = \frac{\omega_1^H}{\omega_n^H} = \pm \frac{z_2 \ldots z_k}{z_1 \ldots z_{k-1}}$$

计算周转轮系传动比

$$i_{1k} = \frac{\omega_1}{\omega_k} \qquad i_{1H} = \frac{\omega_1}{\omega_H}$$

7 复合轮系传动比计算





$-\omega_H$

周转轮系　　假想的定轴轮系

$-\omega_{H1}$

假想的定轴轮系　　新周转轮系

复合轮系传动比计算（计算思路）

在计算混合轮系传动比时，既不能将整个轮系作为定轴轮系来处理，也不能对整个机构采用转化机构的办法。

?

A. 首先将各个基本轮系正确地区分开来.

B. 分别列出计算各基本轮系传动比的方程式.

C. 找出各基本轮系之间的联系.

D. 将各基本轮系传动比方程式联立求解，即可求得混合轮系的传动比.

复合轮系传动比计算（计算思路）

基本轮系的划分

行星轮 → 系杆 → 中心轮 → 周转轮系1

行星轮 → 系杆 → 中心轮 → 周转轮系1

•

•

•

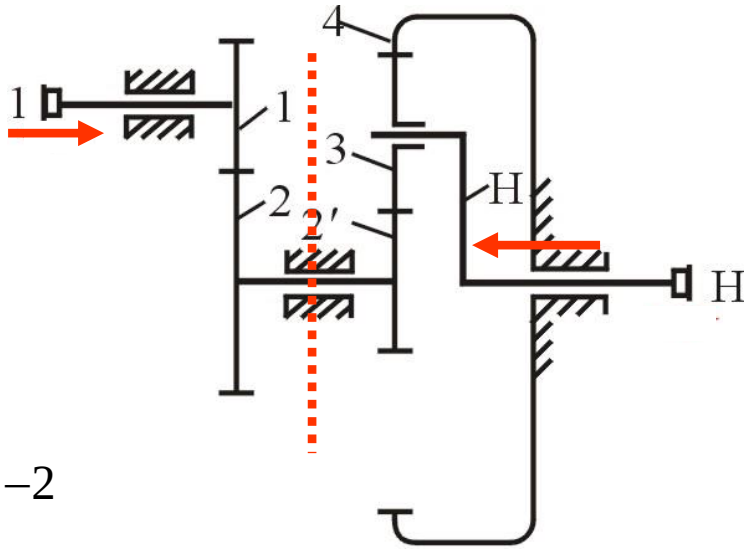
行星轮 → 系杆 → 中心轮 → 周转轮系n

定轴轮系

复合轮系传动比计算（计算举例）

例题3： 已知 $z_1 = 20$，$z_2 = 40$，$z_2' = 20$，$z_3 = 80$，

试求传动比 i_{1H}。



①分析轮系的组成

周转轮系 $2' - 3 - 4 - H$

定轮轮系 $1 - 2$

②分别写出各基本轮系的传动比

定轮轮系 $$i_{12} = \frac{n_1}{n_2} = -\frac{Z_2}{Z_1} = -\frac{40}{20} = -2$$

周转轮系 $$i_{2'4}^{H} = \frac{n_{2'} - n_H}{n_4 - n_H} = 1 - \frac{n_{2'}}{n_H} = -\frac{Z_4}{Z_{2'}} = -\frac{80}{20}$$

③两个轮系之间的关系 $n_2 = n_{2'}$

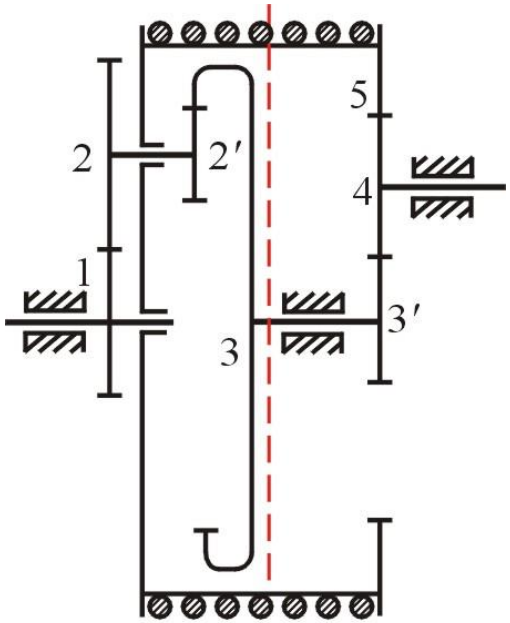
④联立求解 $$i_{1H} = \frac{n_1}{n_H} = i_{12} \cdot i_{2'H} = -10$$ 齿轮1和系杆H转向相反

复合轮系传动比计算（计算举例）

例题4：电动卷扬机减速器中，

已知 $z_1=24$，$z_2=52$，$z_2'=21$ $z_{3'}=18$，$z_5=78$

试求传动比 $i_{1\mathrm{H}}$。



①分析轮系的组成

周转轮系 $1-(2-2')-3-5(H)$

定轮轮系 $3'-4-5$

②分别写出各基本轮系的传动比

周转轮系 $$i_{13}^{\mathrm{H}}=\frac{\omega_1-\omega_H}{\omega_3-\omega_H}=-\frac{z_2}{z_1}\frac{z_3}{z_{2'}}$$

定轮轮系 $$i_{3'5}=\frac{\omega_3}{\omega_5}=-\frac{z_4}{z_{3'}}\frac{z_5}{z_4}=-\frac{z_5}{z_{3'}}$$

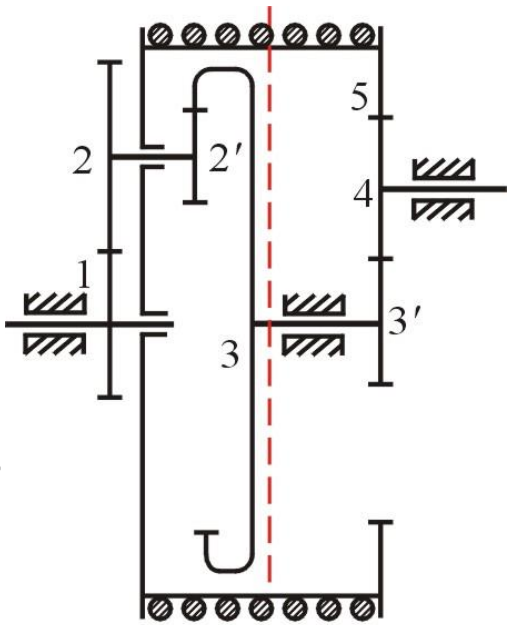
复合轮系传动比计算（计算举例）

③两个轮系之间的关系

$$\omega_5 = \omega_H$$

④联立求解

$$i_{15} = \frac{\omega_1}{\omega_5} = 1 + \frac{z_2 z_3}{z_1 z_{2'}} + \frac{z_5 z_2 z_3}{z_{3'} z_1 z_{2'}} = 54.3$$



齿轮1与齿轮5的转向相同